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LLM-COORDINATION: DEVELOPING COORDINATING AGENTS WITH LARGE LANGUAGE MODELS

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Abstract

LLM-Coordination: Developing Coordinating Agents with Large Language Models

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It is essential for intelligent agents to not only excel in isolated situations but also coordinate with partners to achieve common goals. Current Multi-agent Coordination methods rely on Reinforcement Learning techniques to train agents that can work together effectively. On the other hand, agents based on Large Language Models (LLM) have shown promising reasoning and planning capabilities in single-agent tasks, at times outperforming RL-based methods. In this study, we build and assess the effectiveness of LLM agents in various coordination scenarios. We introduce the LLM-Coordination Framework to enable LLMs to complete coordination tasks. We evaluate our method on three game environments and organize the evaluation into five aspects: Theory of Mind, Situated Reasoning, Sustained Coordination, Robustness to Partners, and Explicit Assistance. First, the evaluation of the Theory of Mind and Situated Reasoning reveals the capabilities of LLM to infer the partner's intention and reason actions accordingly. Then, the evaluation around Sustained Coordination and Robustness to Partners further showcases the ability of LLMs to coordinate with an unknown partner in complex long-horizon tasks, outperforming Reinforcement Learning baselines. Lastly, to test Explicit Assistance, which refers to the ability of an agent to offer help proactively, we introduce

two novel layouts into the Overcooked-AI benchmark, examining if agents can prioritize helping their partners, sacrificing time that could have been spent on their tasks. This research underscores the promising capabilities of LLMs in sophisticated coordination environments and reveals the potential of LLMs in building strong real-world agents for multi-agent coordination.

To My Parents and my Partner.

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Chapter 1

Introduction

The need for coordination can arise from mundane tasks like cooking and cleaning to more essential tasks like coordinated search-and-rescue operations. The assimilation of AI tools and agents into the human experience cannot be complete without AI agents being able to coordinate in a similar manner. Current methods for Multi-Agent coordination predominantly utilize Reinforcement Learning approaches that suffer from issues of sample efficiency, lack of robustness to new partners, and lack of interpretability. More Recently, Agents based on Large Language Models have emerged as a new paradigm for developing situated, task-oriented agents [36, 42]. LLMs are capable of solving reasoning problems with minimal downstream training data due to their pretraining and generate free-text explanations, making them promising candidates for developing coordination agents. While the planning and reasoning abilities of Large Language Models for common-sense and logical reasoning tasks have been extensively studied, their emergent cooperative reasoning abilities are underexplored.

In this study, we introduce the LLM-Coordination Framework to facilitate multi-agent coordination using Large Language Models. We will refer to agents using the LLM-Coordination framework as LLM-Co agents. The LLM-Co framework is an Agent framework with the following features

- Programmatically converts real-time state information to textual description that the LLM can parse.
- Generates feasible action space based on the current state.
- Parses LLM response and converts it to a sequence of low-level actions
- Maintains a memory of previous actions for the LLM

We then systematically evaluate the LLMs and the LLM-Co framework on the following critical competencies required for effective coordination: Theory of Mind Inference (understanding others' beliefs and intentions), Situated Reasoning (contextual action analysis), Sustained Coordination (long-term action planning and adjustment), Robustness to Partners (adapting to new collaborators), and Explicit Assistance (actively aiding partners).

We begin by conducting a preliminary study that allows us to determine the baseline abilities of various Large Language Models and their suitability to be used in the LLM-Coordination Framework. We do this by examining LLMs' Theory of Mind Inference and Situated Reasoning capabilities, which are foundational for coordination. These abilities enable models to understand others' intentions and contextualize these understandings within their environment. We introduce the "LLM-ToM-Reasoning Test Set," comprising diverse scenarios from multiple coordination environments, to assess LLMs' performance in deducing optimal actions based on their partner's intentions and environmental context. Comparing four different LLMs (GPT-4, GPT-3.5-turbo, Vicuna-33B, and Vicuna-13B), we find GPT-4 to be significantly superior, getting an almost perfect score. The other LLMs cannot provide sufficient correct responses to be suitable for following multi-turn coordination.

Next, we evaluate the sustained coordination abilities of LLM-Coordination Agents. We use GPT-4 as the LLM of choice as it is the only candidate that provides acceptable ToM and Situated Reasoning scores. Providing the LLM with Task Description, Turn-by-turn state information, action memory, and a list of feasible actions at each timestep, we explore the sustained coordination reasoning in LLMs on the Overcooked-AI benchmark. We compare the performance of LLM Agents (w. GPT-4) with Reinforcement Learning (RL) based baselines, which are the gold standards for AI-AI gameplay. We then experiment with varying the partners in the Overcooked Environment. We pair LLM-Co agents with proxy human agents to test the agents' robustness to partners. We observe that LLM agents perform better than or equal to the RL baselines in both AI-AI and AI-human proxy gameplay.

Previous works introducing novel methods for developing Collaborative Agents [46, 21, 45] using LLMs do not test their abilities to understand the concept of common payoff and proactively help their partners (Explicit Assistance). We introduce two new layouts in the Overcooked environment, including a gate element that forces agents to assist their partners in completing deliveries. Through experiments on these new layouts, we discover that LLM agents can determine the right strategy needed to help out their partners. However, they require a "helper prompt" to be attentive to situations where their partner may need such help. We train RL baselines on these new layouts and discover that LLMs with helper directives outperform these baselines.

This research highlights the potential of LLMs in coordination reasoning and contributes a novel perspective to the field of AI-agent collaboration. The LLM-Coordination Framework opens new avenues in the development of AI agents capable of sophisticated, context-aware, and coordinated decision-making, setting a foundation for future explorations in multi-agent AI systems.

1.1 Document Structure

- Chapter 2 will cover some prerequisite concepts and terminology about LLMs, LLM Agents, and Multi-agent Coordination that will be used throughout the document.
- Chapter 3 will discuss prominent related work that explores Large Language Models as agents acting in a situated environment and covers the current state of Multi-agent Coordination Methods and Environments.
- Chapter 4 will introduce the LLM-Coordination Framework to enable LLMs to complete coordination games.
- Chapter 5 will cover the environments we have used to conduct experiments, in-